Lecture 3: Section 0.4

Inverse Functions

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Definition of Inverse Functions

Inverse Function Definition

If f is a function, then its inverse f^{-1} is a function such that:

$$f^{-1}(f(x)) = x \quad \text{for all } x \text{ in domain of } f$$

$$f(f^{-1}(x)) = x \quad \text{for all } x \text{ in domain of } f^{-1}$$

Notation

- Inverse of f is denoted by f^{-1}
- Read as "f inverse"
- **Important**: $f^{-1}(x) \neq \frac{1}{f(x)}$

Example

Example

Verify if functions are inverses:

1. $f(x) = 2x + 3$ and $g(x) = \frac{x-3}{2}$
2. $f(x) = x^2$ and $g(x) = \sqrt{x}$
3. $f(x) = \frac{1}{x}$ and $g(x) = \frac{1}{x}$

Domain and Range of Inverse Functions

Key Property

- Domain of f = Range of f^{-1}
- Range of f = Domain of f^{-1}

Example

Find domain and range of f and f^{-1}:

1. $f(x) = 2x + 1$
2. $f(x) = \sqrt{x - 3}$
3. $f(x) = \frac{1}{x-2}$
4. $f(x) = x^2$ (for $x \geq 0$)
5. $f(x) = e^x$

Method for Finding Inverse Functions

Step-by-Step Method

1. Replace $f(x)$ with y
2. Interchange x and y
3. Solve for y
4. Replace y with $f^{-1}(x)$
5. Verify domain and range

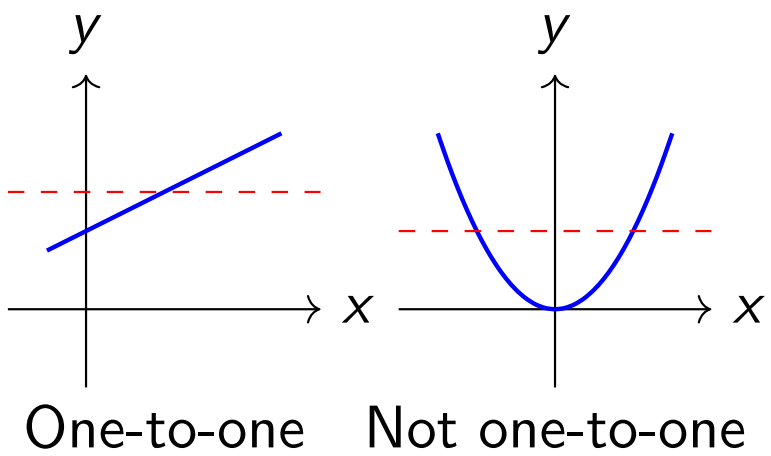
Find the inverse of each function:

1. $f(x) = 3x - 7$
2. $f(x) = \frac{2x+1}{x-3}$
3. $f(x) = \sqrt{x+4}$
4. $f(x) = x^3 + 2$
5. $f(x) = e^{2x-1}$

Existence of Inverse Functions

Horizontal Line Test

A function has an inverse **if and only if** it is **one-to-one** (passes the horizontal line test).



Existence of Inverse Functions

Example

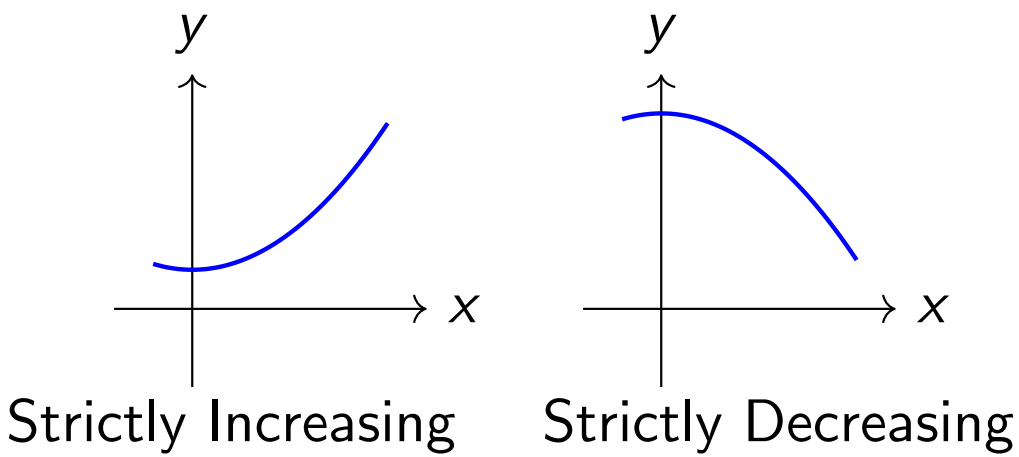
Determine which functions have inverses:

1. $f(x) = x^3$
2. $f(x) = x^2$
3. $f(x) = \sin(x)$
4. $f(x) = 2^x$
5. $f(x) = |x|$

Increasing or Decreasing Functions are Invertible

Theorem

If a function is strictly increasing or strictly decreasing on its domain, then it is one-to-one and therefore invertible.



Increasing or Decreasing Functions

Example

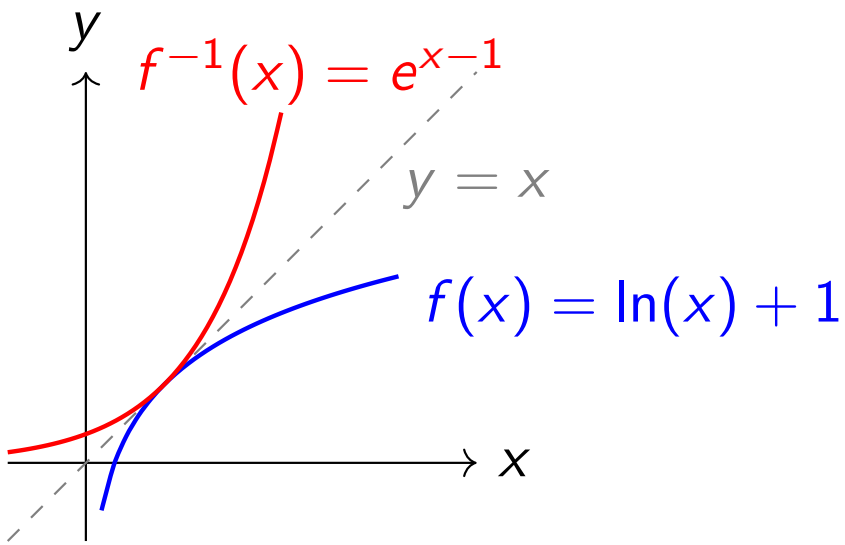
Show these functions are invertible by proving they're monotonic:

1. $f(x) = 3x - 5$
2. $f(x) = e^{-x}$
3. $f(x) = \ln(x)$ for $x > 0$
4. $f(x) = \frac{1}{x}$ for $x \neq 0$

Graphs of Inverse Functions

Key Property

The graph of f^{-1} is the reflection of the graph of f across the line $y = x$.



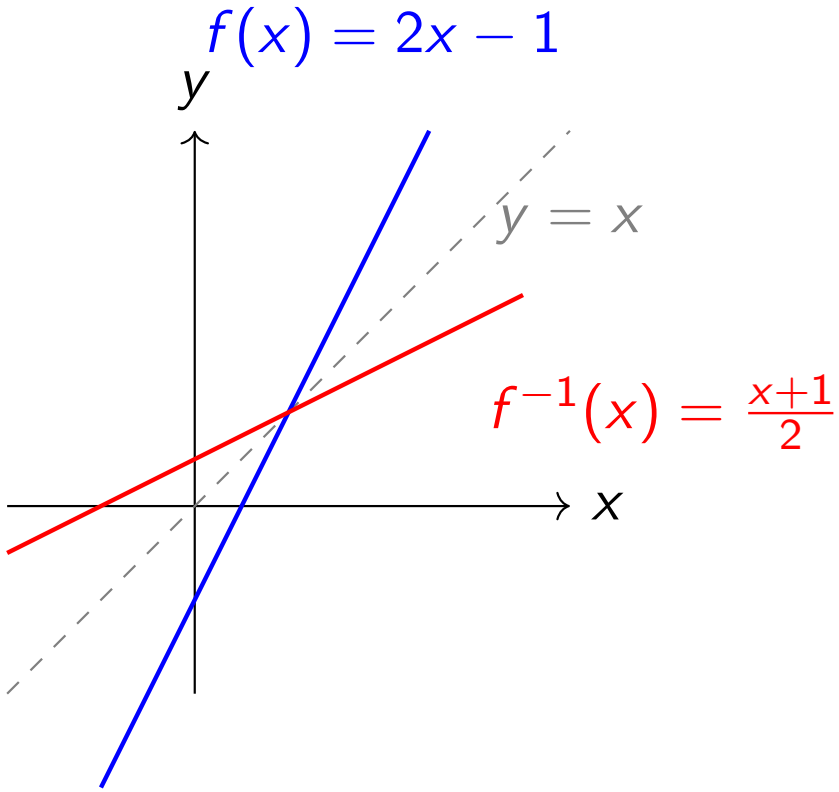
Graphs of Inverse Functions

Example

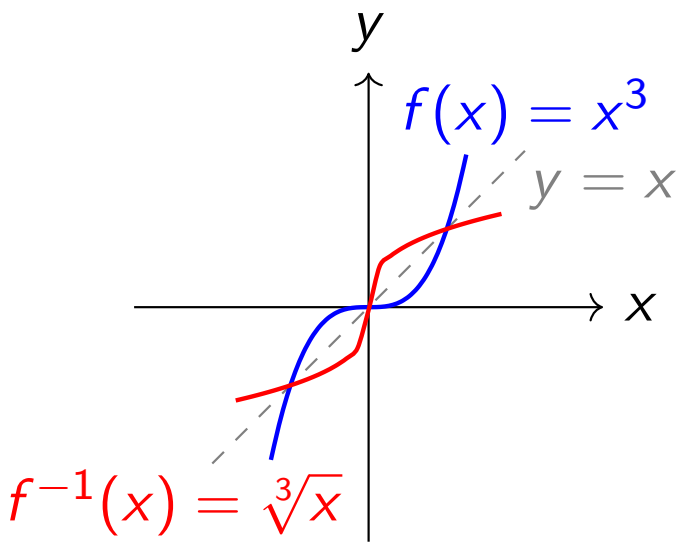
Sketch graphs of f and f^{-1} for:

1. $f(x) = 2x - 1$
2. $f(x) = x^3$
3. $f(x) = \sqrt{x}$
4. $f(x) = \frac{1}{x}$

Graph: $f(x) = 2x - 1$ and its Inverse



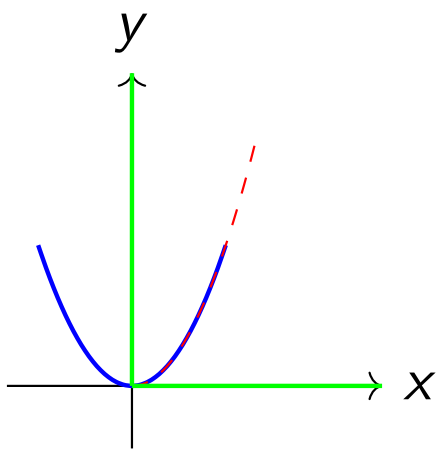
Graph: $f(x) = x^3$ and its Inverse



Restricting Domains for Invertibility

Strategy

If a function fails the horizontal line test, we can restrict its domain to make it one-to-one.



Restrict to $x \geq 0$ for invertibility

Restricting Domains for Invertibility

Example

Restrict domains to make functions invertible:

1. $f(x) = x^2$
2. $f(x) = \sin(x)$
3. $f(x) = (x-2)^2 + 1$
4. $f(x) = |x|$
5. $f(x) = x^4$

Practice Problems: Basic Concepts

Definition and Verification

1. Verify that $f(x) = \frac{x-1}{3}$ and $g(x) = 3x + 1$ are inverses
2. Determine if $f(x) = x^2 + 1$ (for $x \geq 0$) and $g(x) = \sqrt{x-1}$ are inverses
3. Find the domain and range of $f(x) = \sqrt{4-x}$ and its inverse

Finding Inverses

1. Find the inverse of $f(x) = \frac{3x-2}{x+1}$
2. Find the inverse of $f(x) = 2 + \sqrt{x-3}$

Practice Problems: Advanced Concepts

Existence and Graphs

1. Determine which functions have inverses without restricting domains:
   - $f(x) = x^3 - 3x$
   - $f(x) = e^x + e^{-x}$
   - $f(x) = \frac{1}{x^2}$
2. Sketch the graph of $f(x) = (x-2)^2$ and its inverse after restricting the domain
3. Prove that $f(x) = \ln(x + \sqrt{x^2+1})$ is its own inverse

Practice Problems: Advanced Concepts

Applications

1. The function $C(f) = \frac{5}{9}(f - 32)$ converts Fahrenheit to Celsius. Find its inverse and interpret

2. If $f(x) = 2^x$ gives bacterial growth, what does $f^{-1}(x)$ represent?

Key Properties and Summary

Essential Properties

- $f^{-1}(f(x)) = x$ and $f(f^{-1}(x)) = x$
- Domain of f = Range of f^{-1}
- Range of f = Domain of f^{-1}
- Graph of f^{-1} is reflection across $y = x$
- A function has an inverse iff it's one-to-one

Methods Summary

- Use horizontal line test to check for invertibility
- To find inverse: swap x and y, then solve for y
- Restrict domain if function is not one-to-one
- Verify inverse by composition

Inverse Trigonometric Functions

Definition

Inverse trigonometric functions are the inverses of trigonometric functions with restricted domains to make them one-to-one.

Principal Inverse Trigonometric Functions

- $\arcsin(x)$ or $\sin^{-1}(x)$: Inverse of $\sin(x)$ on $[-\frac{\pi}{2}, \frac{\pi}{2}]$
- $\arccos(x)$ or $\cos^{-1}(x)$: Inverse of $\cos(x)$ on $[0, \pi]$
- $\arctan(x)$ or $\tan^{-1}(x)$: Inverse of $\tan(x)$ on $(-\frac{\pi}{2}, \frac{\pi}{2})$
- $\operatorname{arccot}(x)$: Inverse of $\cot(x)$ on $(0, \pi)$
- $\operatorname{arcsec}(x)$: Inverse of $\sec(x)$ on $[0, \pi]$, $x \neq \frac{\pi}{2}$
- $\operatorname{arccsc}(x)$: Inverse of $\csc(x)$ on $[-\frac{\pi}{2}, \frac{\pi}{2}]$, $x \neq 0$

Inverse Trigonometric Functions

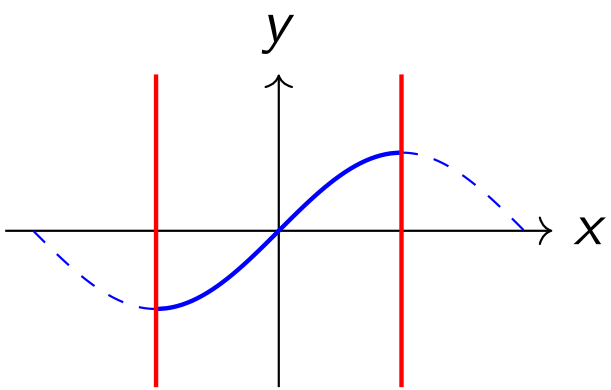
Example

Find the principal values:

1. $\arcsin\left(\frac{1}{2}\right)$
2. $\arccos\left(-\frac{\sqrt{2}}{2}\right)$
3. $\arctan(1)$
4. $\operatorname{arccot}(-\sqrt{3})$

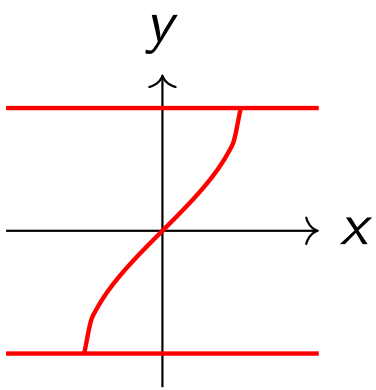
Graph: $y = \sin(x)$ and $y = \arcsin(x)$

$y = \sin(x)$



Restricted domain: $[-\frac{\pi}{2}, \frac{\pi}{2}]$

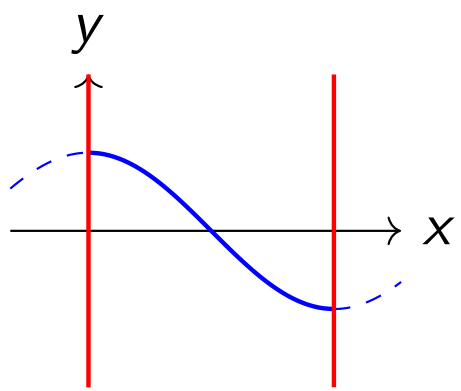
$y = \arcsin(x)$



Range: $[-\frac{\pi}{2}, \frac{\pi}{2}]$

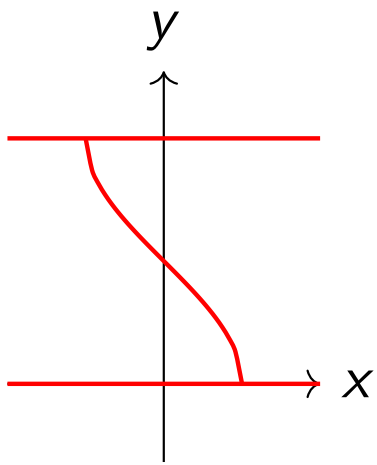
Graph: $y = \cos(x)$ and $y = \arccos(x)$

$y = \cos(x)$



Restricted domain: $[0, \pi]$

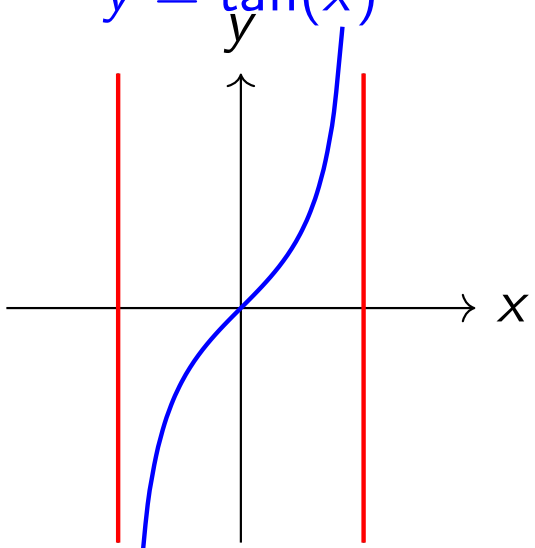
$y = \arccos(x)$



Range: $[0, \pi]$

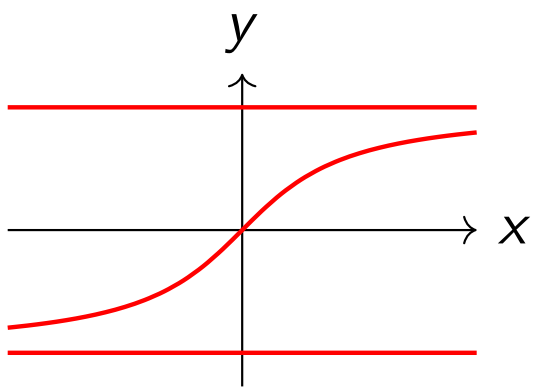
Graph: $y = \tan(x)$ and $y = \arctan(x)$

$y = \tan(x)$



Restricted domain: $(-\frac{\pi}{2}, \frac{\pi}{2})$

$y = \arctan(x)$



Range: $(-\frac{\pi}{2}, \frac{\pi}{2})$

Evaluating Inverse Trigonometric Functions

Key Concepts

- Inverse trig functions return angles
- Results are always in the principal range
- Use reference angles and quadrant analysis

Example

Evaluate each expression:

1. $\arcsin\left(\frac{\sqrt{3}}{2}\right)$
2. $\arccos\left(-\frac{1}{2}\right)$
3. $\arctan(-\sqrt{3})$
4. $\sin^{-1}\left(\sin\left(\frac{5\pi}{6}\right)\right)$
5. $\cos^{-1}\left(\cos\left(\frac{4\pi}{3}\right)\right)$
6. $\tan^{-1}\left(\tan\left(\frac{3\pi}{4}\right)\right)$

Inverse Trigonometric Functions

Composition Rules

- $\sin^{-1}(\sin x) = x$ only if $x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$
- $\cos^{-1}(\cos x) = x$ only if $x \in [0, \pi]$
- $\tan^{-1}(\tan x) = x$ only if $x \in (-\frac{\pi}{2}, \frac{\pi}{2})$

Identities for Inverse Trigonometric Functions

Fundamental Identities

- $\sin^{-1}(x) + \cos^{-1}(x) = \frac{\pi}{2}$ for $-1 \leq x \leq 1$
- $\tan^{-1}(x) + \cot^{-1}(x) = \frac{\pi}{2}$ for all real x
- $\sec^{-1}(x) + \csc^{-1}(x) = \frac{\pi}{2}$ for $|x| \geq 1$

Negative Argument Identities

- $\sin^{-1}(-x) = -\sin^{-1}(x)$
- $\cos^{-1}(-x) = \pi - \cos^{-1}(x)$
- $\tan^{-1}(-x) = -\tan^{-1}(x)$

Identities of Inverse Trigonometric Functions

Example

Prove the identities:

1. $\cos\left(\sin^{-1}(x)\right) = \sqrt{1-x^2}$
2. $\sin\left(\cos^{-1}(x)\right) = \sqrt{1-x^2}$
3. $\tan\left(\sin^{-1}(x)\right) = \frac{x}{\sqrt{1-x^2}}$

Example

Verify the identities:

1. $\sin^{-1}\left(\frac{1}{2}\right) + \cos^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{2}$
2. $\cos^{-1}\left(-\frac{\sqrt{2}}{2}\right) = \pi - \cos^{-1}\left(\frac{\sqrt{2}}{2}\right)$
3. $\tan^{-1}(-1) = -\tan^{-1}(1)$

Practice Problems: Inverse Trigonometry

Evaluation and Composition

1. Evaluate: $\arcsin\left(-\frac{\sqrt{2}}{2}\right) + \arccos\left(-\frac{\sqrt{2}}{2}\right)$
2. Find the exact value: $\tan^{-1}(\sqrt{3}) + \cot^{-1}(-\sqrt{3})$
3. Simplify: $\sin^{-1}\left(\sin\left(\frac{2\pi}{3}\right)\right)$
4. Prove: $\cos^{-1}(-x) = \pi - \cos^{-1}(x)$ for $-1 \leq x \leq 1$

Key Features and Summary

Inverse Trigonometric Functions

- Restricted domains ensure one-to-one property
- Return angle values in specific ranges
- Useful for solving trigonometric equations
- Have important identities and composition properties

Conclusion

Common Inverse Pairs

- $f(x) = e^x$ and $f^{-1}(x) = \ln(x)$
- $f(x) = x^2$ $(x \geq 0)$ and $f^{-1}(x) = \sqrt{x}$
- $f(x) = \sin(x)$ $(-\frac{\pi}{2} \leq x \leq \frac{\pi}{2})$ and $f^{-1}(x) = \arcsin(x)$
- $f(x) = \frac{1}{x}$ and $f^{-1}(x) = \frac{1}{x}$